

Explainable AI

A TECHNICAL PRESENTATION

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Definition

Explainable AI (XAI) is the set of methods and models that make an AI system's decisions understandable to humans.

What is Explainable AI?

- **Interpretability**

How well a human can predict a model's output from its inputs, structure, or parameters.

- **Explainability**

The broader ability to describe, in human terms, why a specific decision was reached.

- **Transparency**

The degree to which a model's internal mechanics are inherently visible, not just approximated after the fact.

Why It Matters



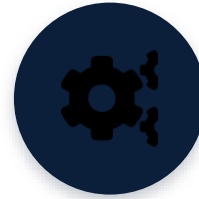
Regulatory

GDPR's "right to explanation" and the EU AI Act require justifiable automated decisions.



Trust

Users and clinicians adopt AI faster when they understand how it reasons.



Debugging

Explanations expose spurious correlations and data leakage before deployment.



Fairness

Attributions reveal whether protected attributes are driving outcomes.

Key Techniques

LIME



Perturbs inputs near a single sample and fits a simple surrogate model to approximate the black box locally.

SHAP



Uses cooperative game theory to fairly attribute a prediction's outcome across all input features.

Grad-CAM



Highlights the image regions that most influenced a convolutional network's classification.

Attention



Surfaces which tokens or regions a transformer weighted most heavily when producing its output.

Where XAI Is Deployed Today



Healthcare

Clinicians need to see why a model flags a diagnosis before acting on it.



Finance

Credit and fraud decisions must be explainable to regulators and applicants.



Legal & HR

Risk-scoring and hiring tools face direct scrutiny for bias and due process.



Autonomous Systems

Engineers must trace why a vehicle or robot made a safety-critical choice.



Open Challenges

Explainability methods are powerful, but not without real limitations.

Limitations & Trade-offs

- **Fidelity gaps**

A surrogate explanation can misrepresent the true model — approximations are not guarantees.

- **Instability**

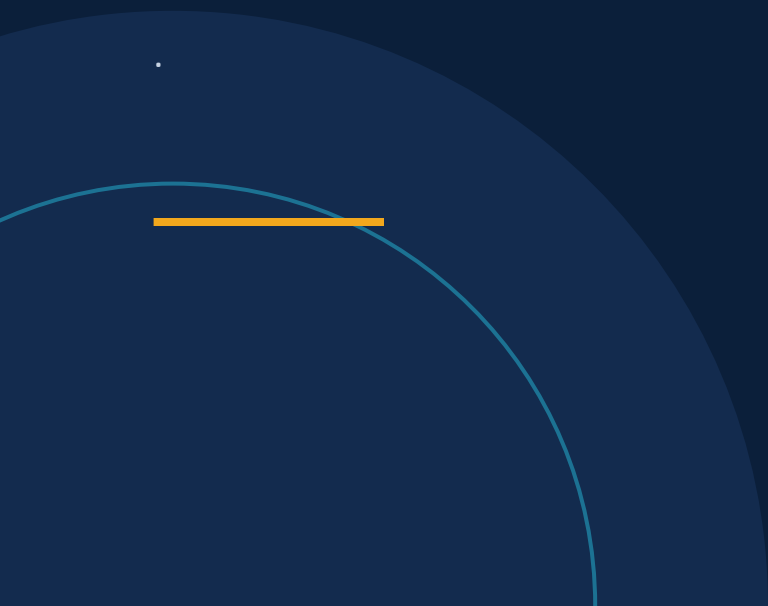
Small input perturbations can produce noticeably different LIME or SHAP explanations.

- **Computational cost**

Exact Shapley values scale exponentially with feature count; most tools approximate.

- **Adversarial manipulation**

Explanations themselves can be gamed to look fair while the underlying model is not.



Thank you

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GPS Map Camera

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